等厚干涉

“光学”模块的知识结构

● 细化的“课程知识点”

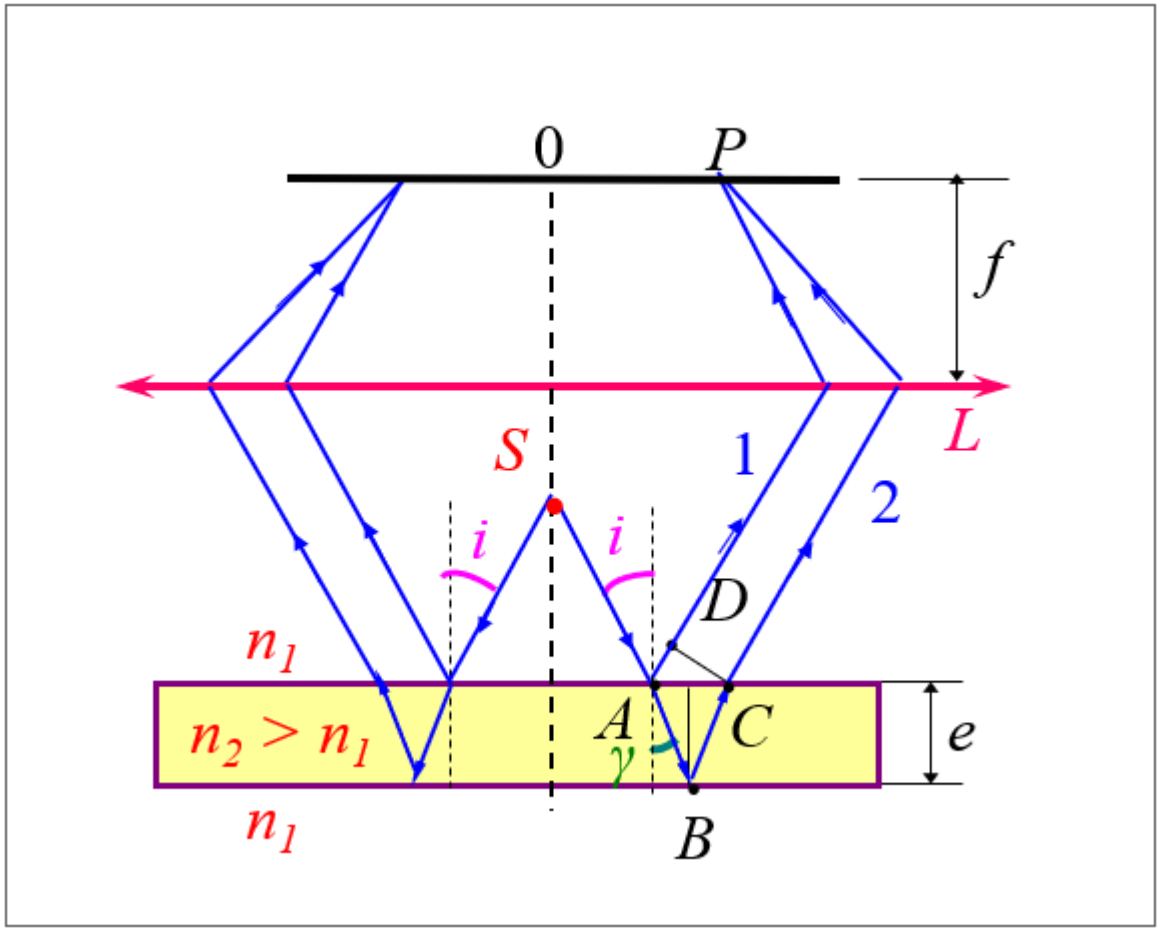
- 几何光学基本定律（教材11.14节）、光程、费马原理、薄透镜成像（1次课）
- 相干光、光程差、杨氏双缝干涉；菲涅尔双镜*、劳埃德镜*（1次课）
- 薄膜干涉、等倾干涉；等厚干涉：劈尖干涉、牛顿环、迈克尔孙干涉仪*（2次课）
- 惠更斯-菲涅尔原理，单缝夫琅禾费衍射、半波带法；圆孔衍射、光学仪器的分辨本领(2次课)
- 衍射光栅、光栅方程、光栅衍射的光强分布、缺级现象(1次课)
- 光的偏振性、偏振片、起偏和检偏、马吕斯定律、反射折射起偏、布儒斯特定律(1次课)
- 晶体双折射现象、惠更斯作图法、偏振棱镜、1/4波片和半波片、偏振光干涉(1次课)

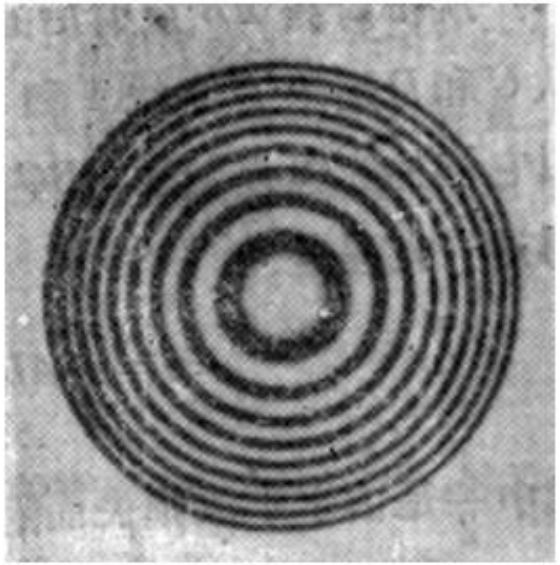
● 主要拓展内容

拓展：大口径光学望远镜及相关技术介绍

拓展：X射线衍射及应用（大物D）

拓展：晶体双折射现象、惠更斯作图法、偏振棱镜、1/4波片和半波片、偏振光干涉





等倾条纹照片

$$\delta = 2n_2 e\cos\gamma + \frac{\lambda}{2}$$

$$\text{或}\delta = 2e\sqrt{n_2^2 - n_1^2\sin^2 i} + \frac{\lambda}{2}$$

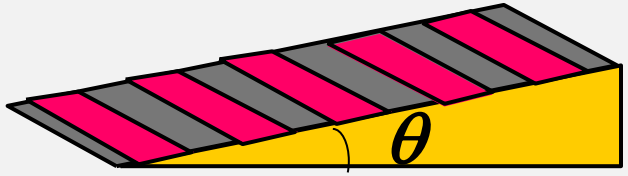
如果薄膜厚度不均匀会怎样?

劈尖(劈形膜)

——夹角很小的两个平面间的劈形薄膜。

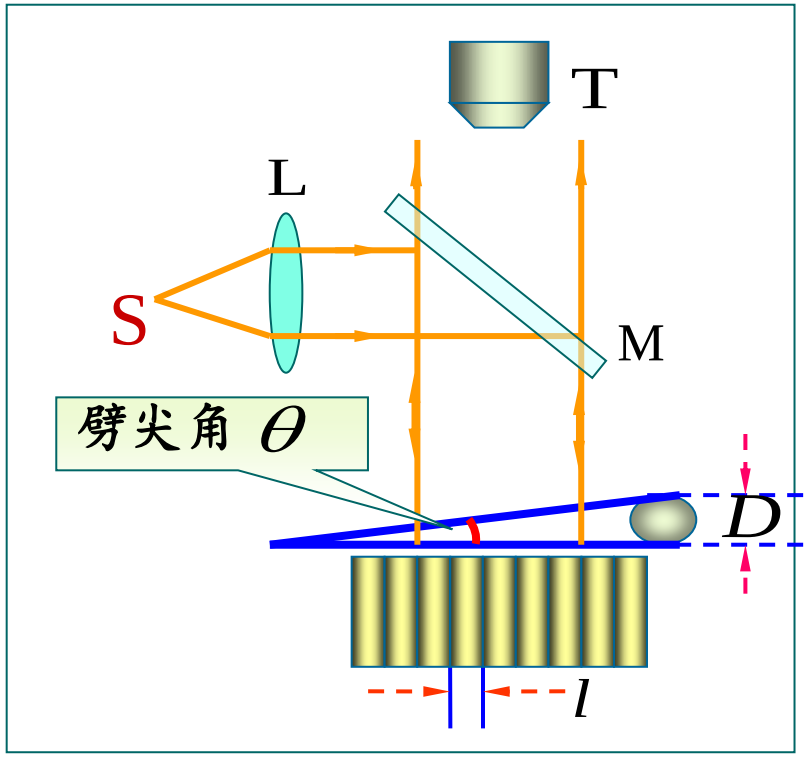
劈尖干涉的实验现象

劈尖干涉在膜表面附近形成平行于交棱的明、暗相间的条纹。



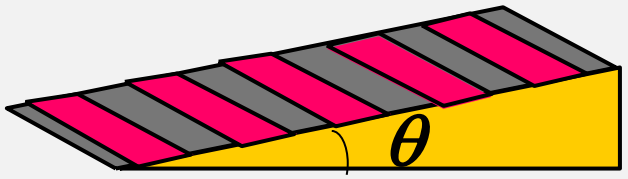
θ：$10^{-4} \sim 10^{-5}$ rad

观察劈尖干涉的实验装置



- **单色平行光垂直入射**
  - 设单色平行光线在A点处入射，膜厚为e，
  - 反射光1, 2叠加要不要考虑半波损失？
- **光程差 ($\theta \to 0$)**

$$\delta = 2ne + \frac{\lambda}{2} \begin{cases} = k\lambda, \quad k = 1,2,3\cdots & \text{亮纹} \\ = (2k+1)\dfrac{\lambda}{2}, \quad k = 0,1,2,\cdots & \text{暗纹} \end{cases}$$



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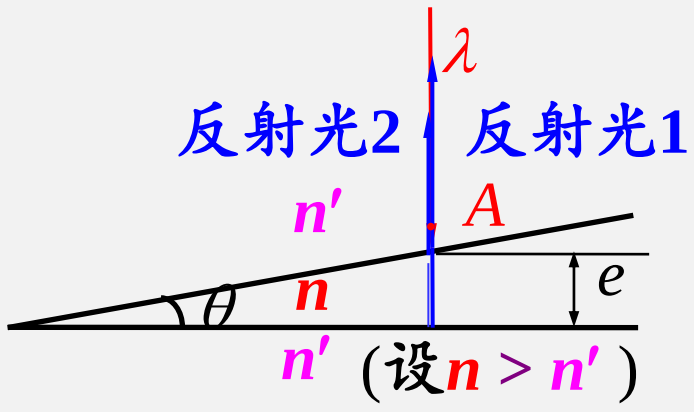
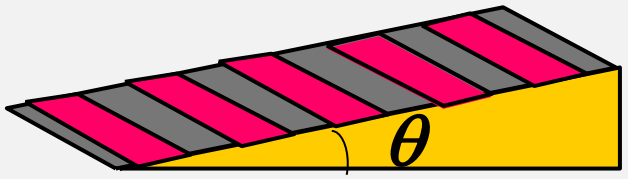


一个厚度e，对应着一个光程差δ，对应着一个条纹——等厚干涉条纹。

$$\delta = 2ne + \frac{\lambda}{2} \begin{cases} = k\lambda, \quad k = 1,2,3\cdots & \text{亮纹} \\ = (2k+1)\dfrac{\lambda}{2}, \quad k = 0,1,2,\cdots & \text{暗纹} \end{cases}$$



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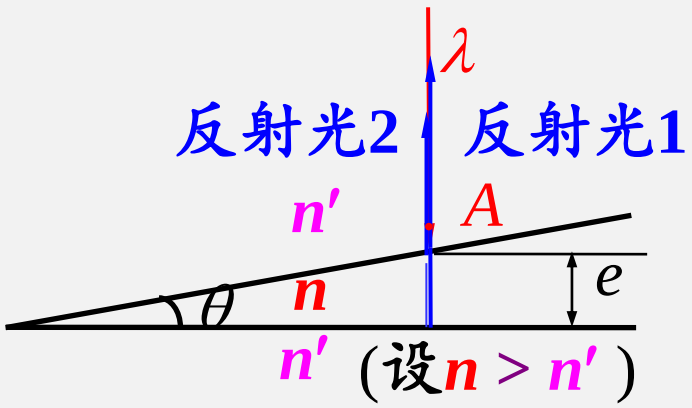


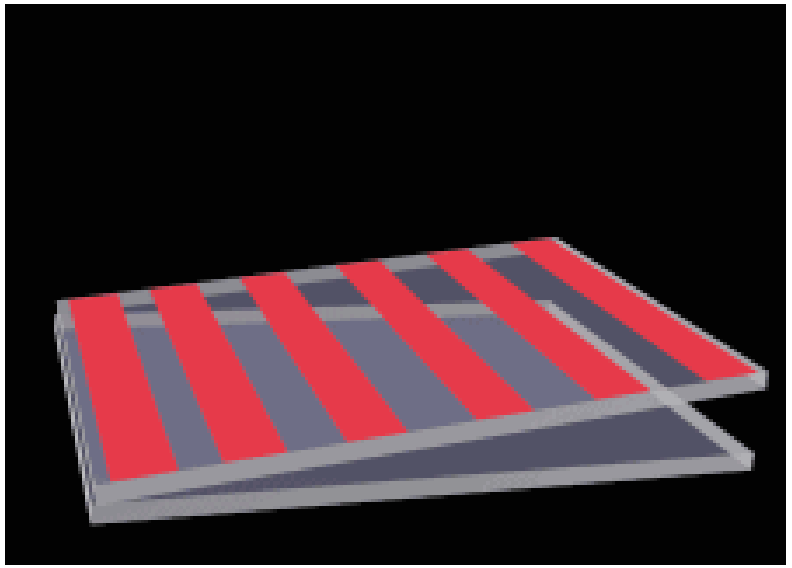
- 棱边处($e = 0$)是暗纹，这是“半波损失”的一例证。
- 亮纹与暗纹等间距地相间排列。
- 相邻两条亮纹对应的厚度差： $\Delta e = \lambda / 2n = \lambda_n / 2.$
- 相邻两条亮纹的间距：

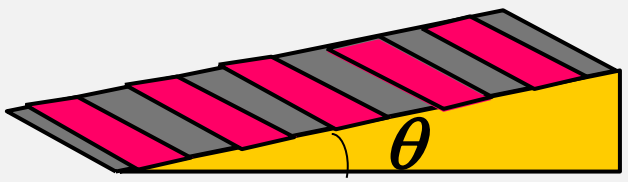
$$L = \frac{\Delta e}{\sin\theta} \approx \frac{1}{\theta}\Delta e = \frac{\lambda}{2n\theta}.$$

当$\theta \to 0$时，干涉条纹在宏观尺度可见。

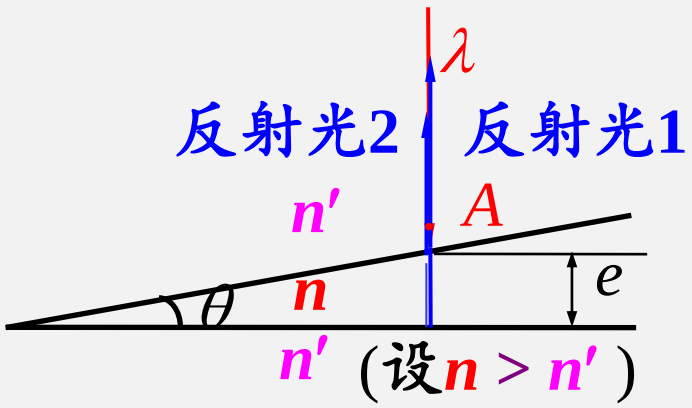
思考1：若夹角变化，条纹怎样?

$$\text{等间隔}\begin{cases}\theta\downarrow,\ \text{变疏}\\ \theta\uparrow,\ \text{变密}\end{cases}$$

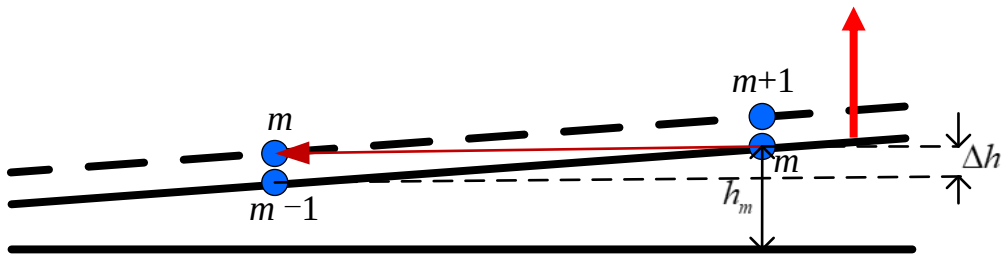




θ：$10^{-4}\sim10^{-5}$ rad

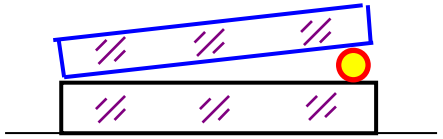


思考2：若劈尖上表面上移，条纹向_____移动。

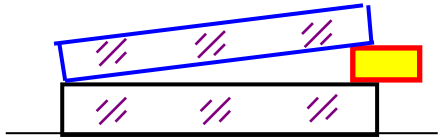


答：条纹向棱边移动（向左侧移动）

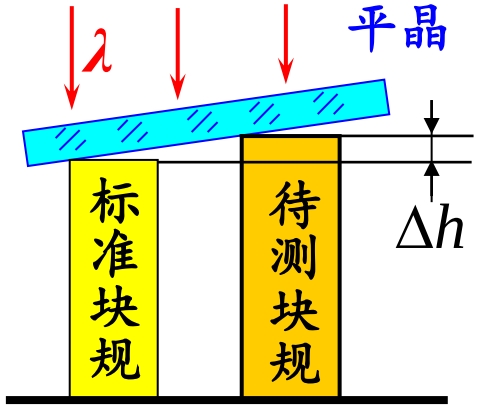
- 测光波的波长λ。 $\delta = 2ne + \dfrac{\lambda}{2}$ $L = \dfrac{\lambda}{2n\theta}.$
- 测微小直径、厚度（或镀膜厚度）。



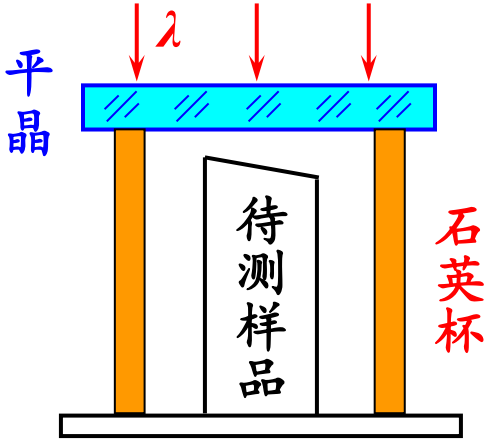
测微小直径



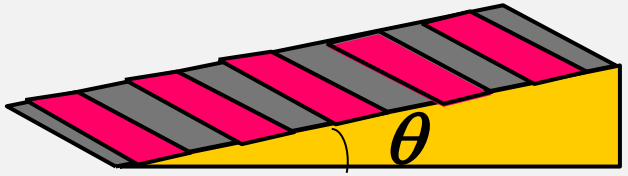
测微小厚度



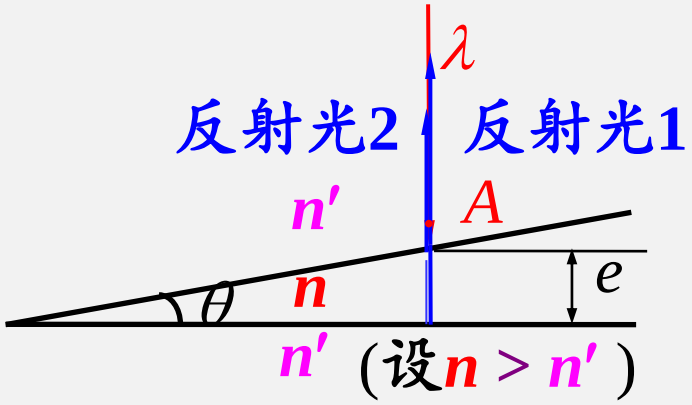
精密测长度



干涉膨胀仪

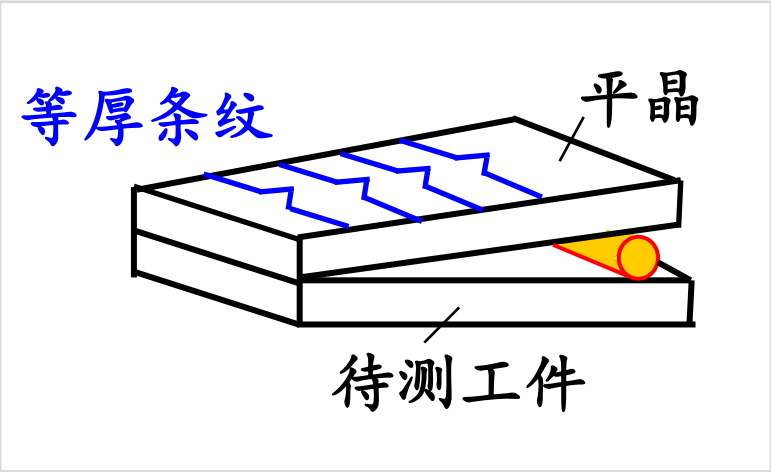


θ : $10^{-4} \sim 10^{-5}$ rad



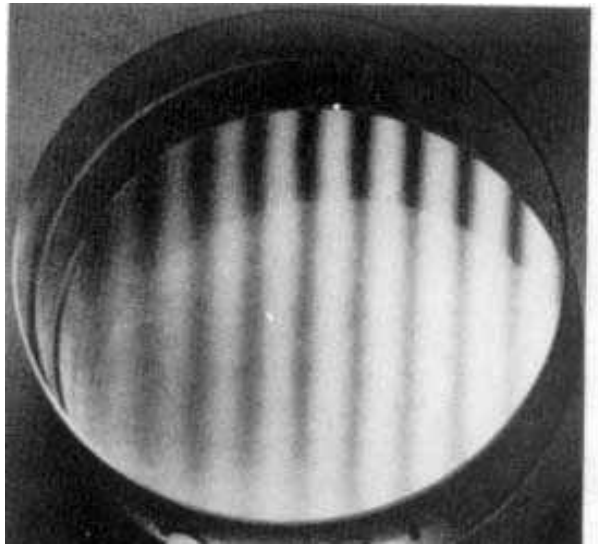
- 测表面平整度。

$$\delta = 2ne + \frac{\lambda}{2} \qquad L = \frac{\lambda}{2n\theta}.$$

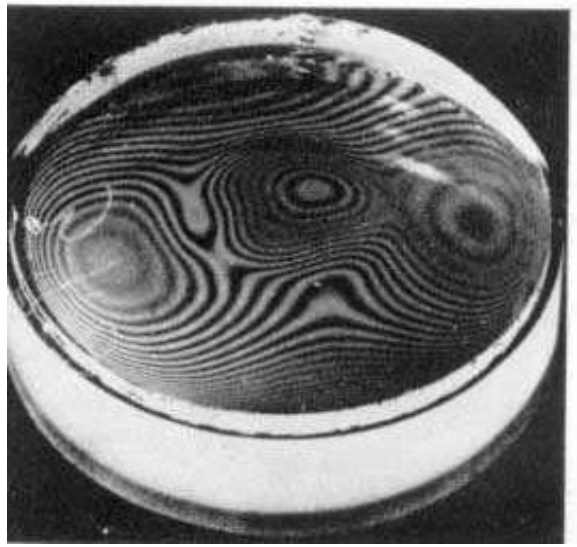


问：右上角图中，待测工件表面是凹纹还是凸纹？

凸纹



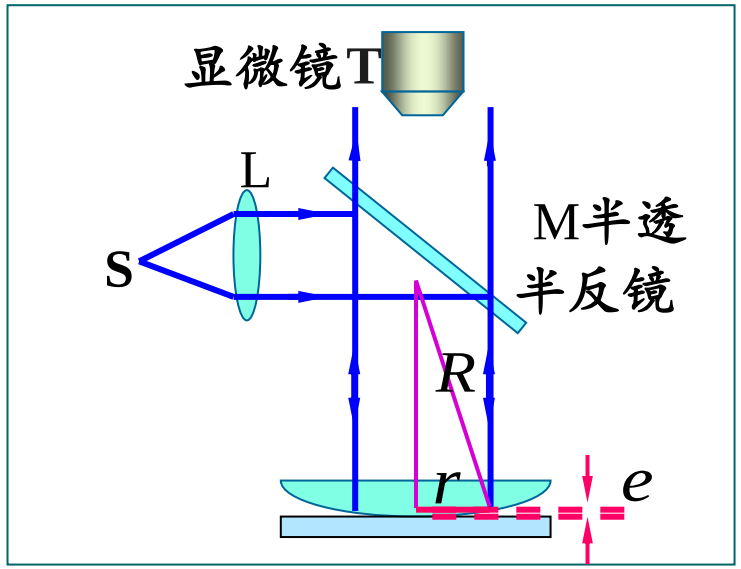
劈尖

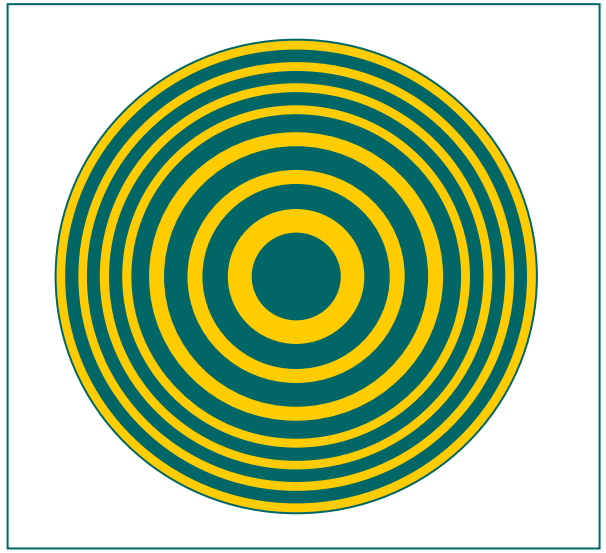


不规则表面

(一) 牛顿环

平凸透镜与平晶间形成空气劈尖。





1. 干涉图样：内疏外密，中心为暗点的圆环。

2. 明纹、暗纹条件：

$$\delta = 2ne + \frac{\lambda}{2} \begin{cases} = k\lambda & \text{明环} \quad k = 1, 2, \cdots \\ = (k + \frac{1}{2})\lambda & \text{暗环} \quad k = 0, 1, 2, \cdots \end{cases}$$

$$\delta = 2ne + \frac{\lambda}{2} \begin{cases} = k\lambda & \text{明环} \quad k = 1, 2, \cdots \\ = (k + \frac{1}{2})\lambda & \text{暗环} \quad k = 0, 1, 2, \cdots \end{cases}$$

3. 明(暗)环的半径:

$$r^2 = R^2 - (R - e)^2 = 2Re - e^2 \approx 2Re$$

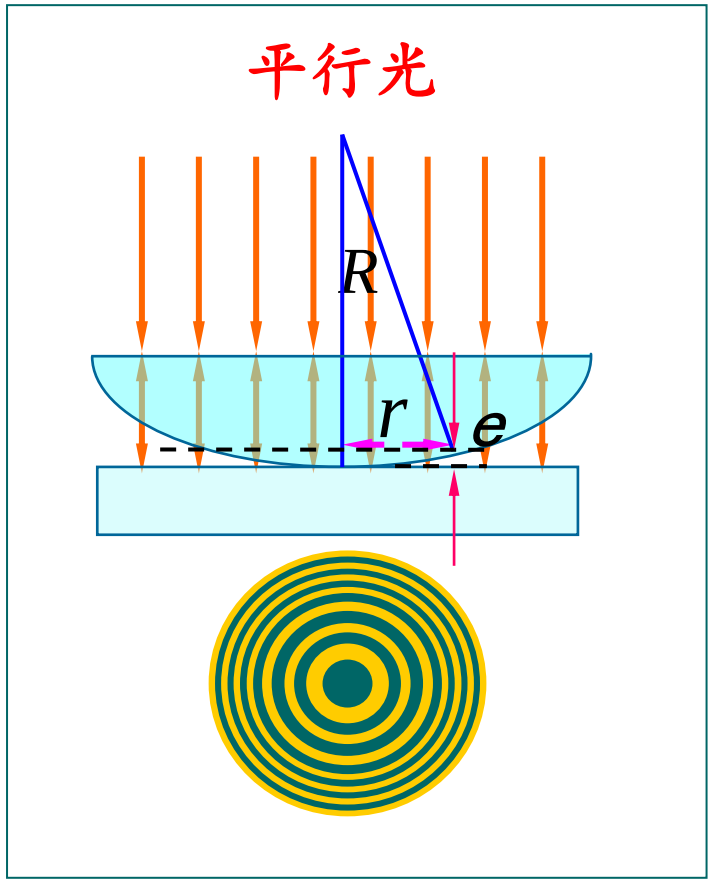
$$\longrightarrow \quad r = \sqrt{2Re}.$$

明环半径

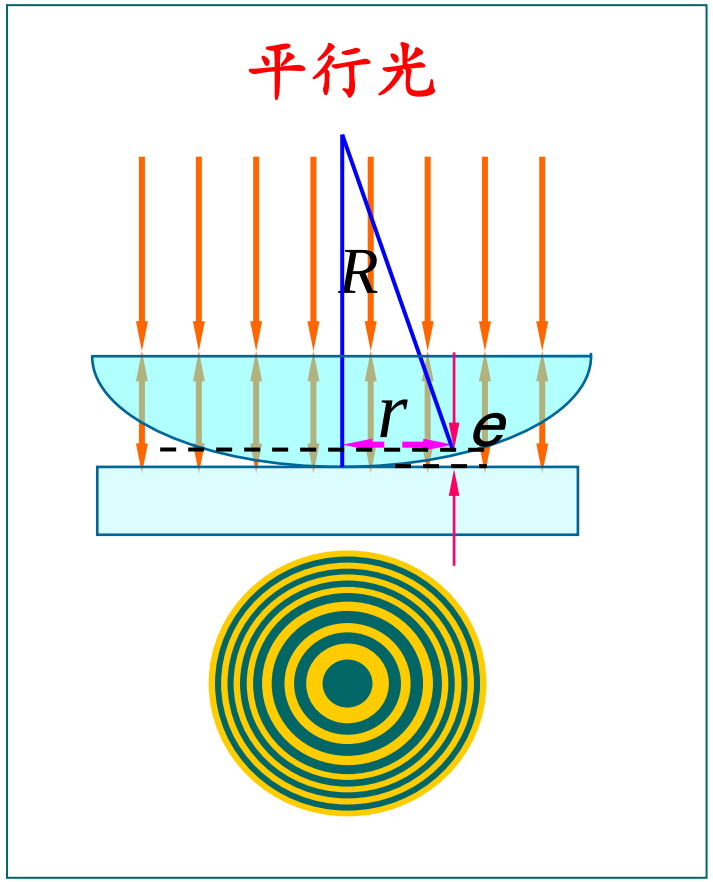
$$r_{明} = \sqrt{\frac{(2k-1)R\lambda}{2n}}. \quad k = 1, 2, \ldots$$

暗环半径

$$r_{暗} = \sqrt{\frac{kR\lambda}{n}}. \quad k = 0, 1, 2, \ldots$$



明环半径 $r_{明}=\sqrt{\dfrac{(2k-1)R\lambda}{2n}}$，暗环半径 $r_{暗}=\sqrt{\dfrac{kR\lambda}{n}}$.



讨论：

1）从反射光中观测，中心点是暗点还是亮点？

从透射光中观测，中心点是暗点还是亮点？

2）条纹间距：

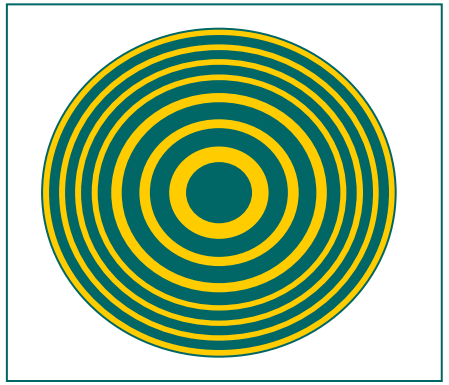
$$r_1 : r_2 : r_3 = 1 : \sqrt{2} : \sqrt{3}.$$

（内疏外密）

3）将牛顿环置于 $n>1$ 的液体中，条纹如何变？

（条纹变密）

明环半径 $r_{明}=\sqrt{\dfrac{(2k-1)R\lambda}{2n}}$, 暗环半径 $r_{暗}=\sqrt{\dfrac{kR\lambda}{n}}$.



牛顿环的应用举例:

1. 测透镜球面的半径 R

已知λ, 数清m, 测出 r_k, r_{k+m}:

$$R=\frac{r_{k+m}^2-r_k^2}{m\lambda}.$$

2. 测入射光的波长:

已知R, 数清 m, 测出 r_k, r_{k+m} :

$$\lambda=\frac{r_{k+m}^2-r_k^2}{mR}.$$

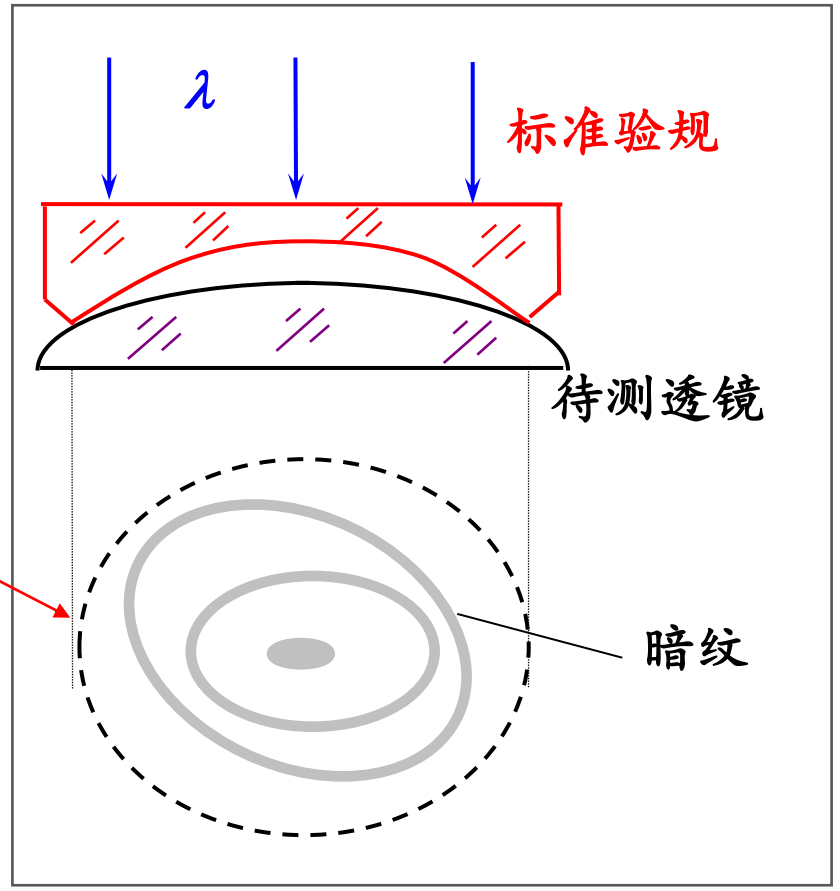
明环半径 $r_{明}=\sqrt{\dfrac{(2k-1)R\lambda}{2n}}$, 暗环半径 $r_{暗}=\sqrt{\dfrac{kR\lambda}{n}}$.

牛顿环的应用举例：

3. 检验透镜（的球面）表面平整度

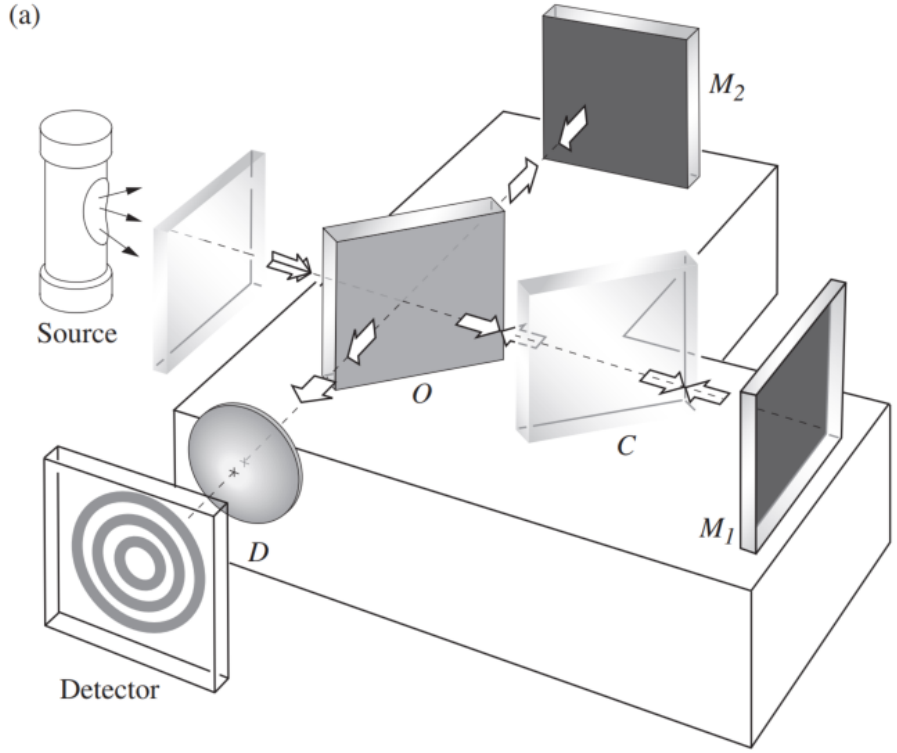
若球面不规则

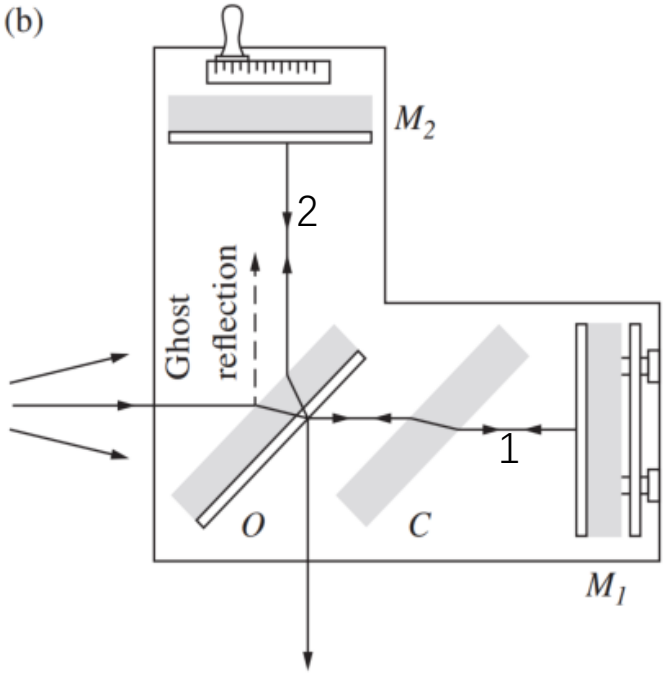
暗纹将不是同心圆环。



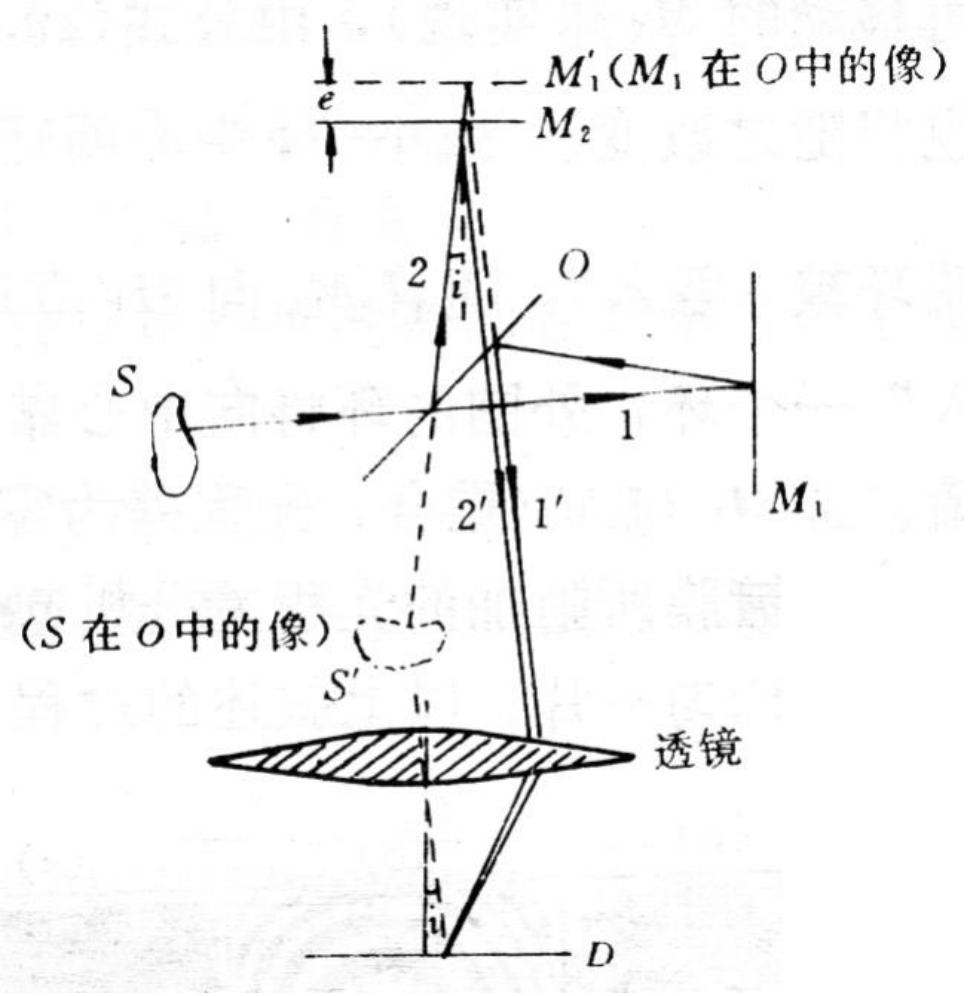
装置与等效光路

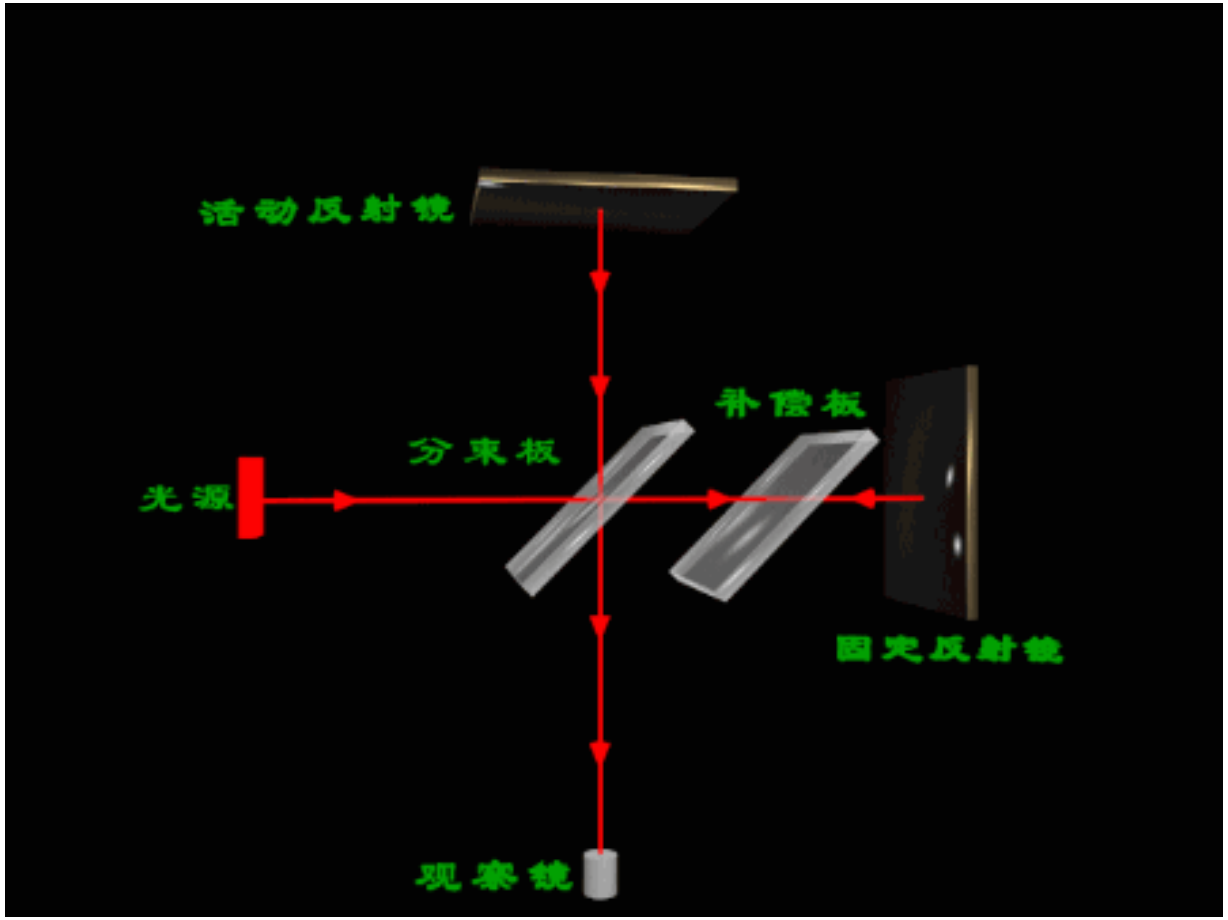
Michelson于1881年设计。迈克耳逊是美国第一个获得诺贝尔物理学奖的学者。





M1固定，M2可移动。 补偿板的作用：除了不镀膜，厚度与分光板一样。①有了补偿板，1和2光线光程差只是几何路径差，（光线1经过3次玻璃板，光线2也一样）；②消除玻璃色散的影响，n=n(λ)，光程与波长有关。有了补偿板，光程差只来源于空气中的几何路径差，用单色性不太好的光源也可以引起干涉。

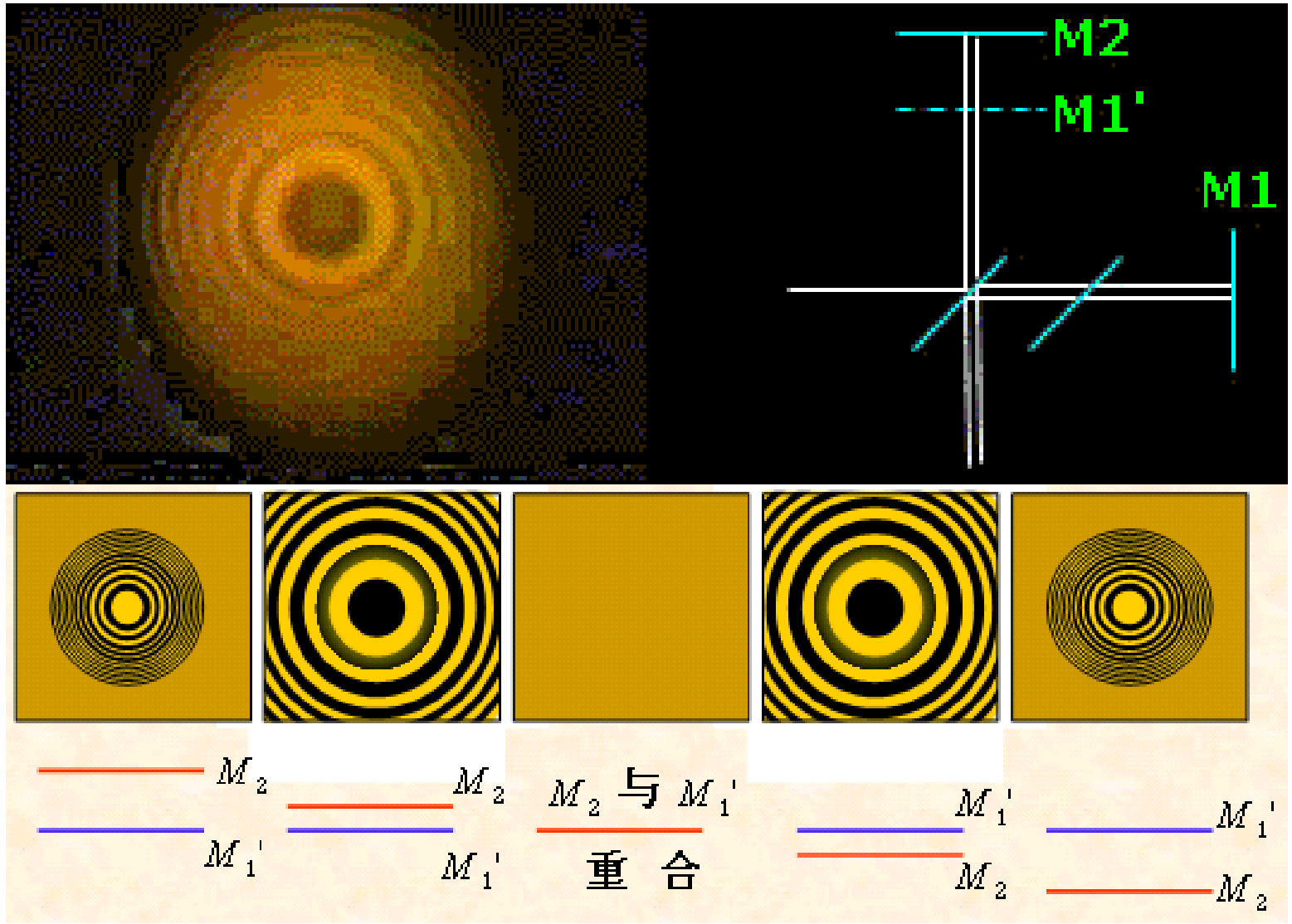




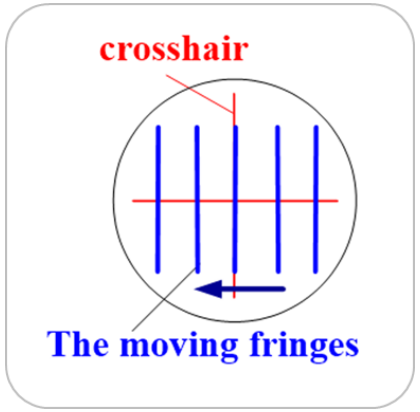
$$\delta = 2h\cos i = m\lambda$$

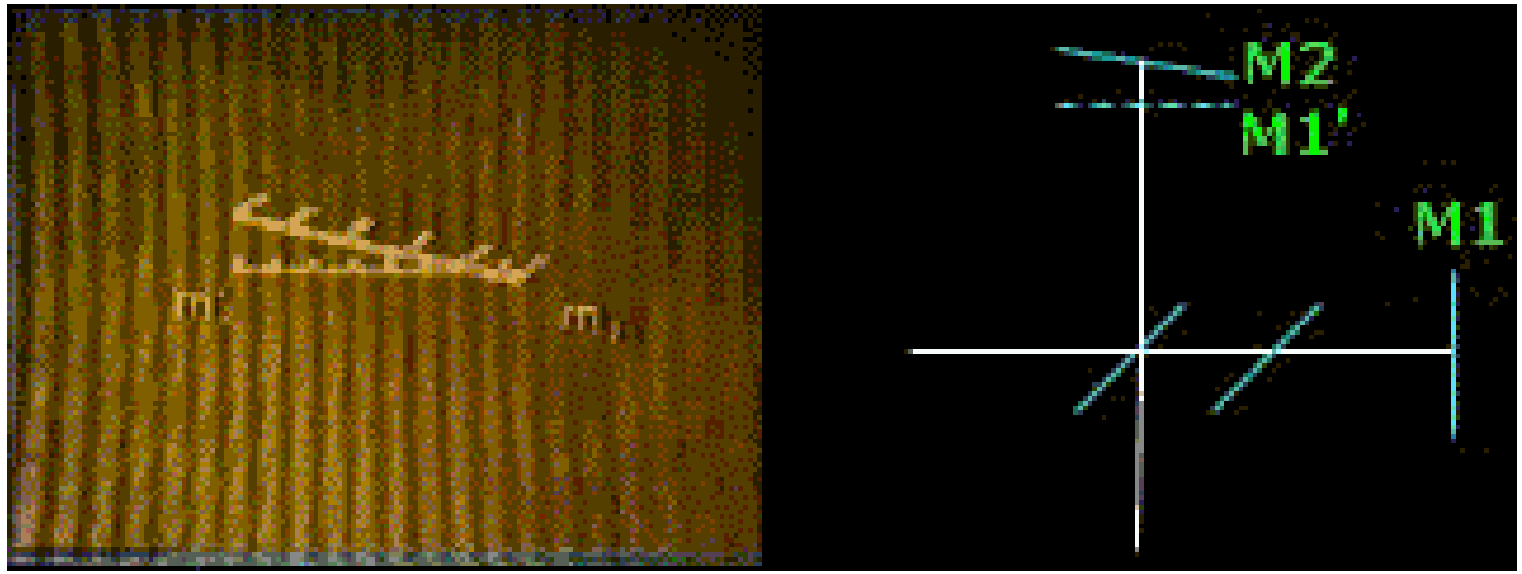
等倾条纹

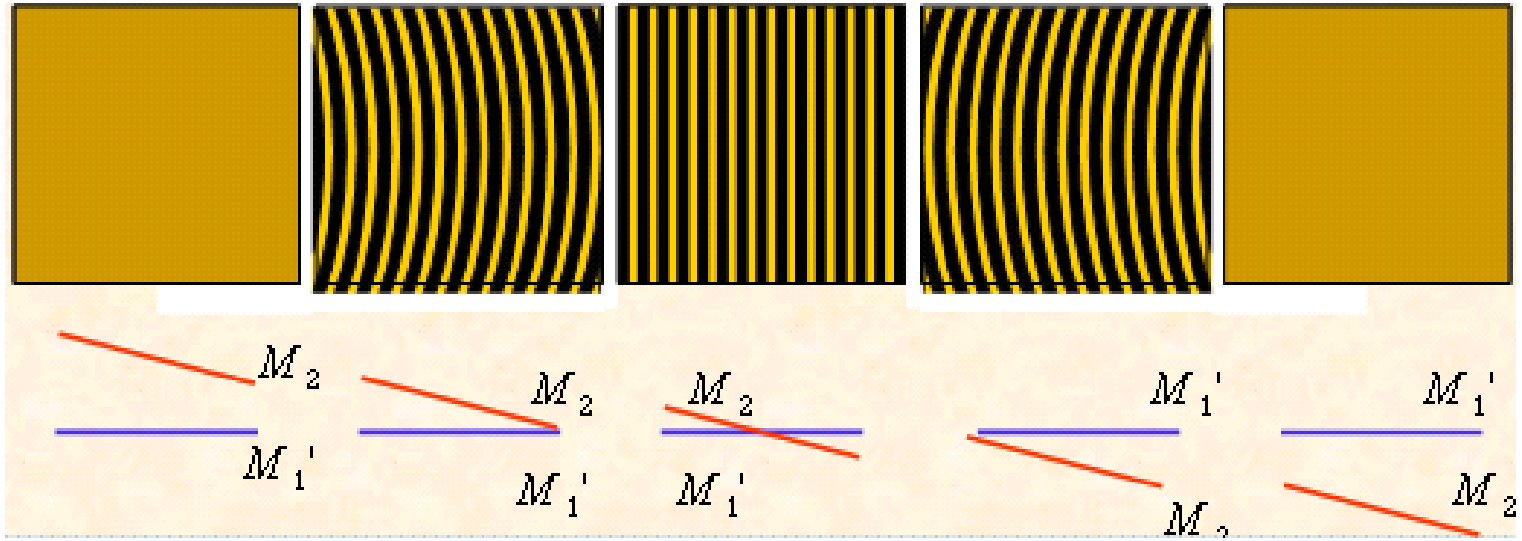
- 中心恰好是m_0级明环： $2h = m_0\lambda$
  向外m_0-1、 m_0-2级次降低
- 若M_2与M'_1相对位置减小，则中心级次减小。h减小一个$\lambda/2$，级次从m_0减小到m_0-1，光斑收缩消失。
- $h\downarrow$， $m_0\downarrow$，视场中环的数目变少，环越来越疏； $h\uparrow$，环越来越密。
- M_2平移过Δh时，条纹有N条缩进或者冒出，则有 $\Delta h = N\dfrac{\lambda}{2}$ 。



等厚条纹

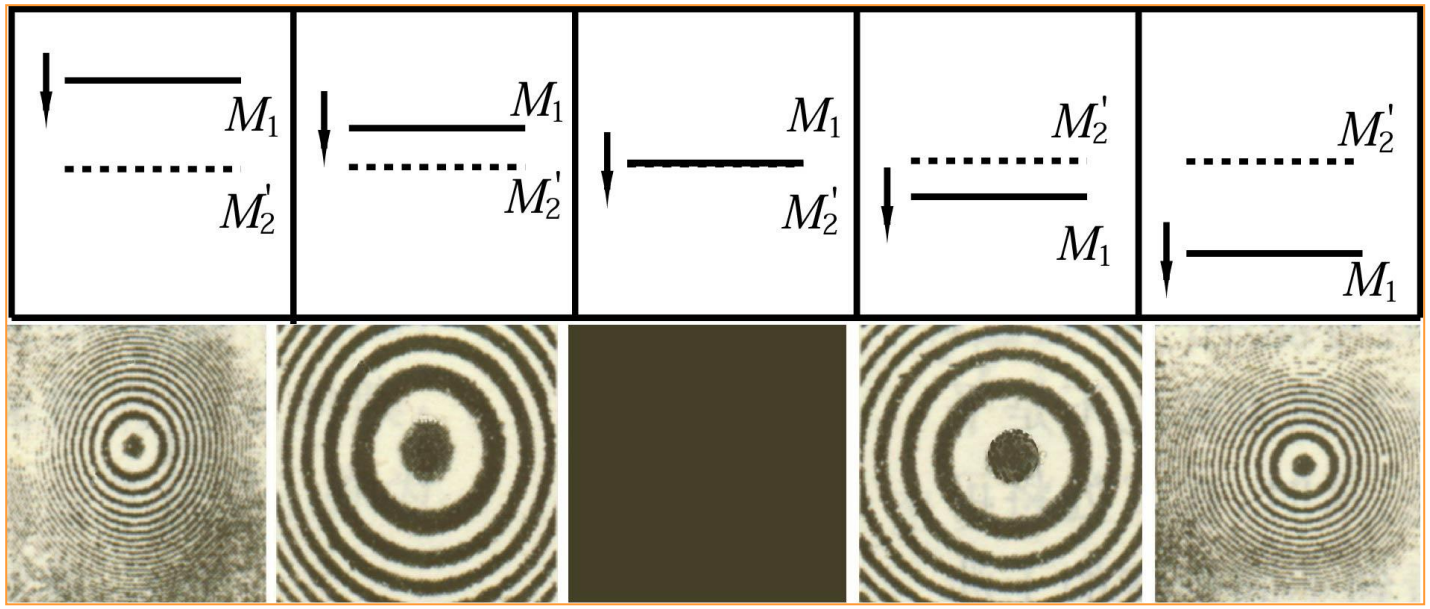




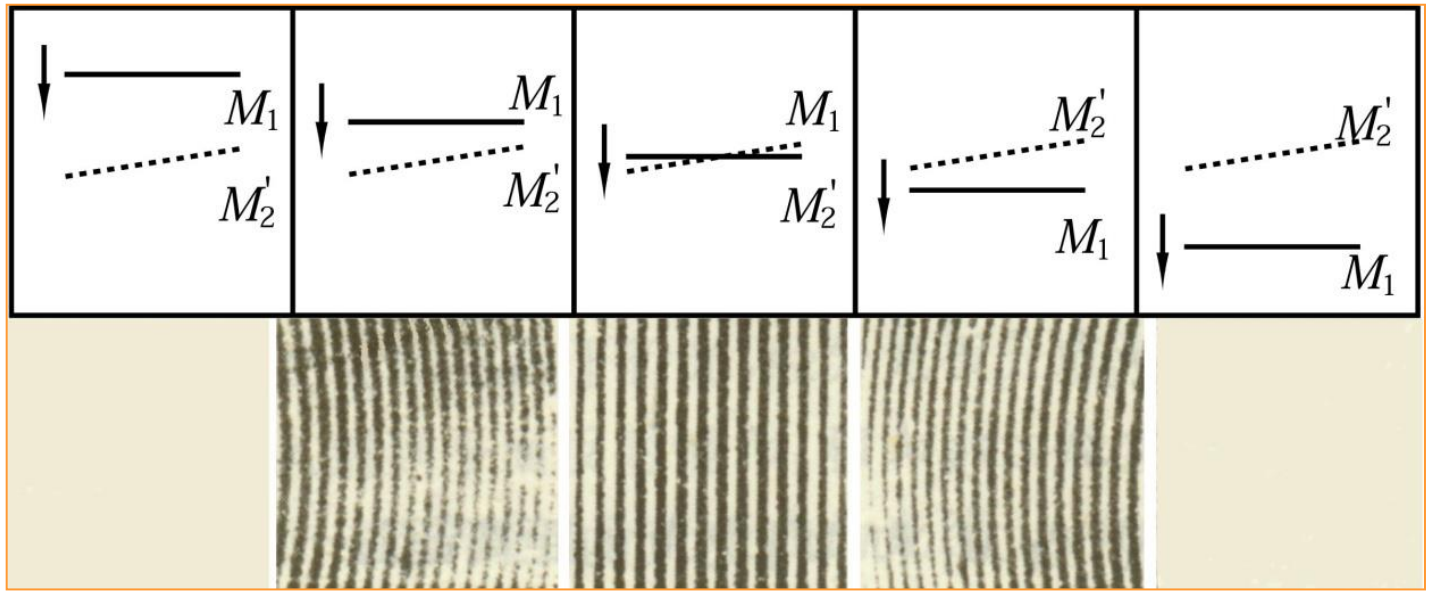


- 视场内设定一条基准线。
- M_2与M'_1相对位置移动一个$\lambda/2$，将有一个条纹移过基准线。
- M_2平移过Δh时，有N条条纹移过基准线，则有 $\Delta h = N\dfrac{\lambda}{2}$ 。

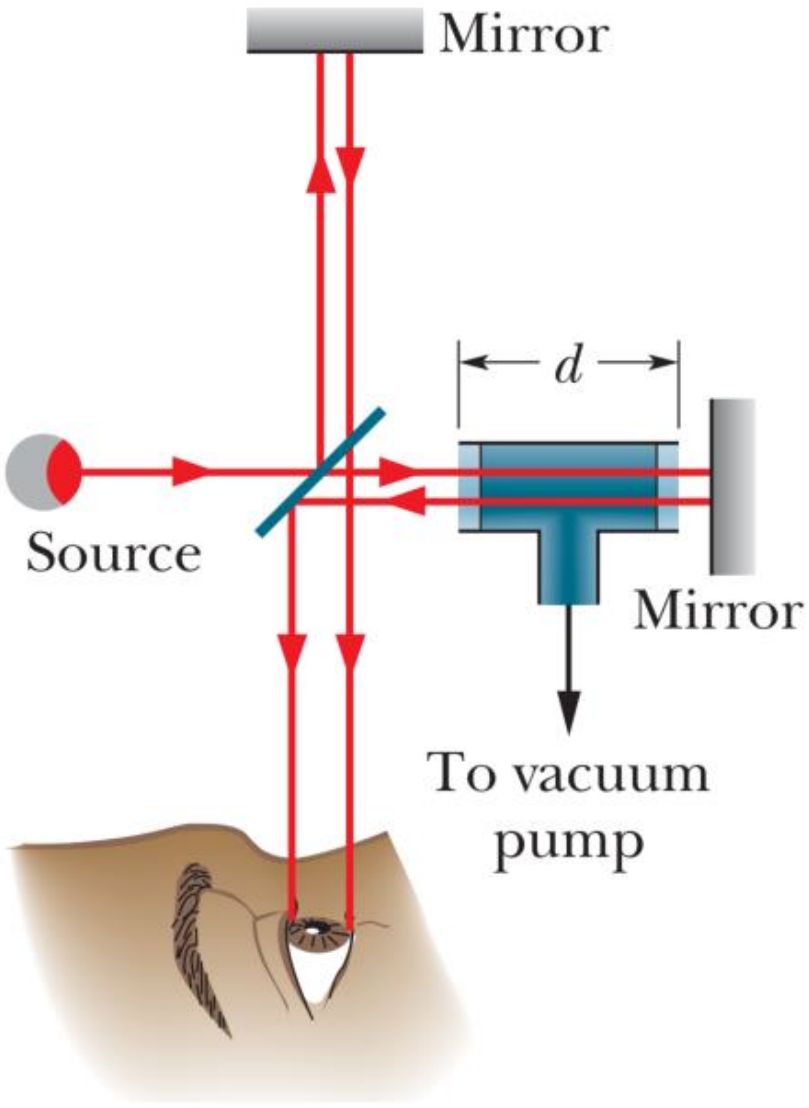
等倾条纹



等厚条纹



例1： 右侧装置可以测量气体折射率。若气室中充满氧气，$d = 10$ cm，$\lambda = 589.3$ nm，抽到真空时，若有92条条纹移过叉丝。求氧气的折射率。



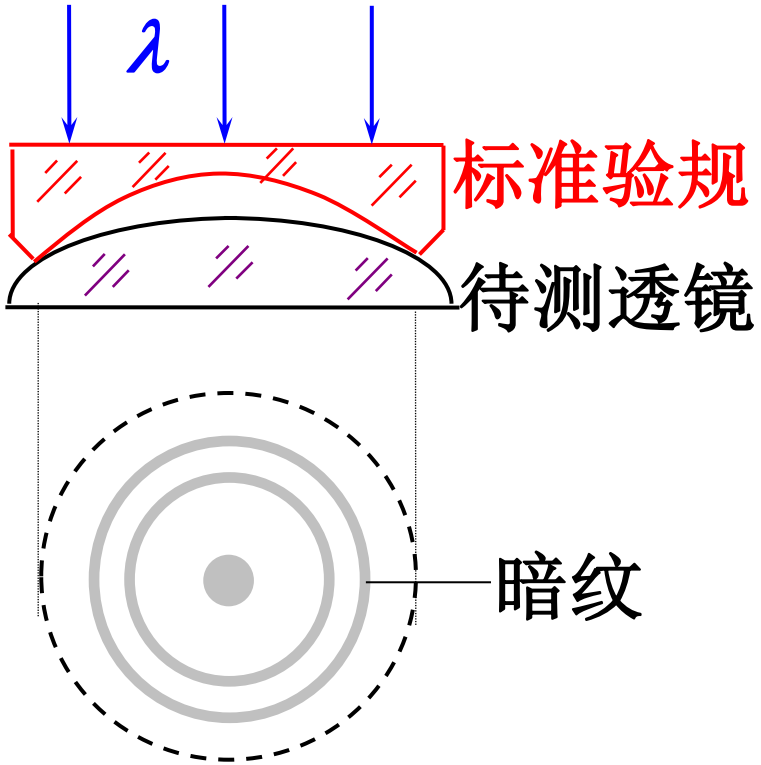
解：光线两次经过气室，光程变化为：

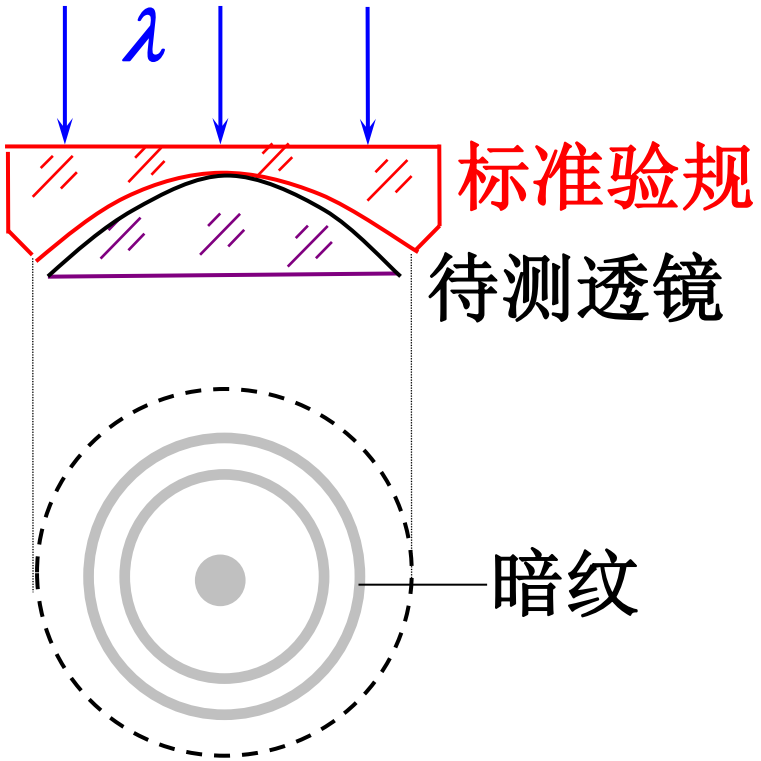
$$\delta = 2(n-1)d = N\lambda$$

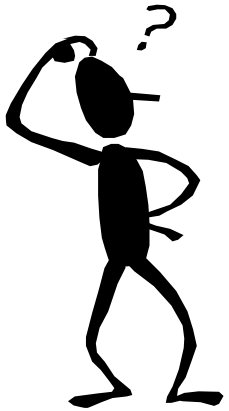
$$n = 1.00000 + \frac{N\lambda}{2d}$$

$$= 1.00000 + \frac{92 \times 589.3 \times 10^{-9}}{2 \times 10^{-2}} = 1.00271$$

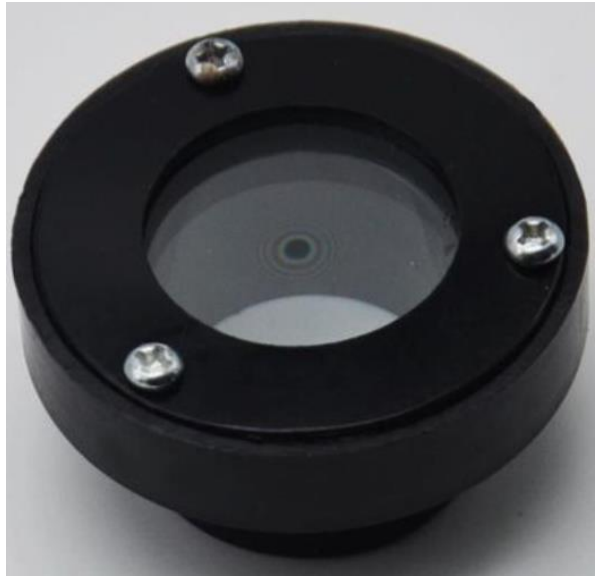
思考题：若两个待测透镜都产生下图所示的暗纹分布，如何区分?

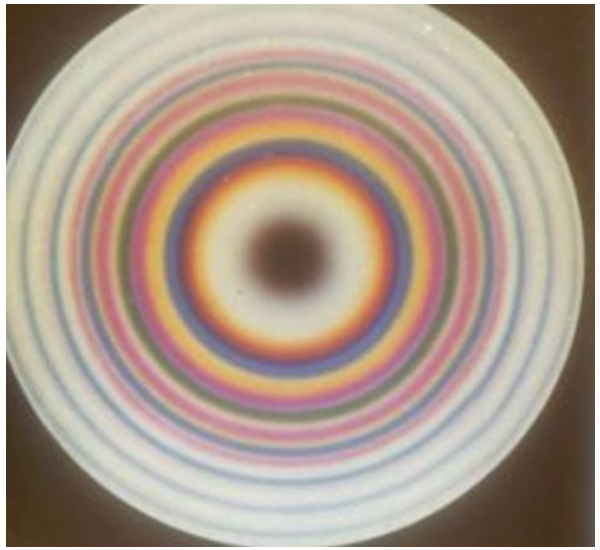


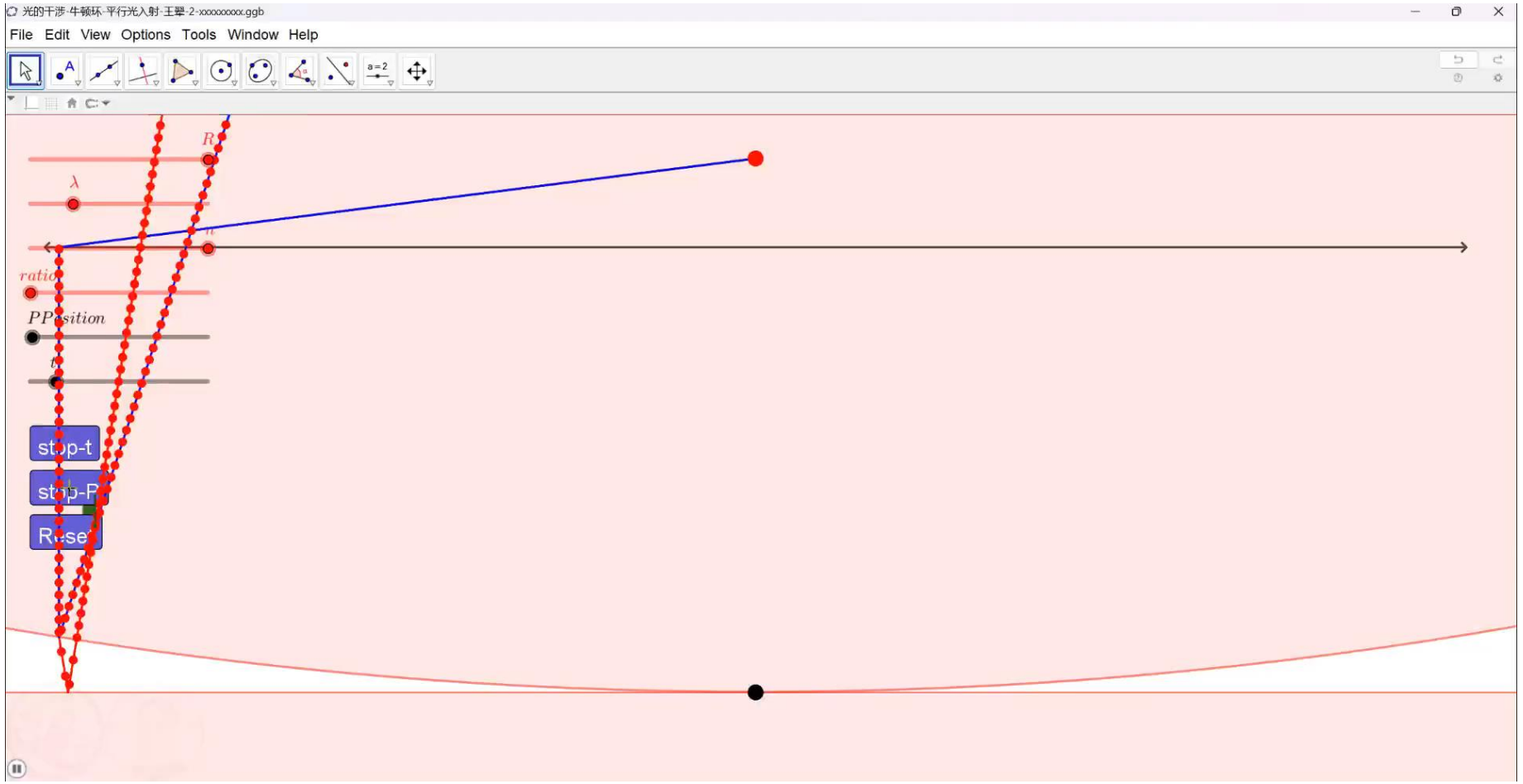




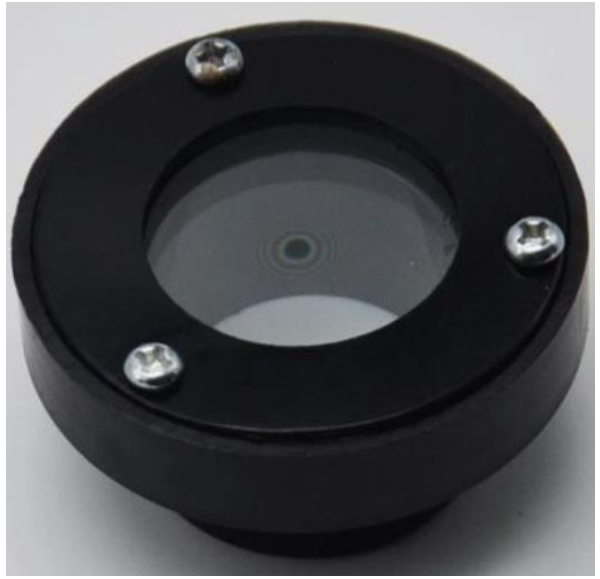
1.1 牛顿环上方的平凸透镜对应的球面半径较大时… …

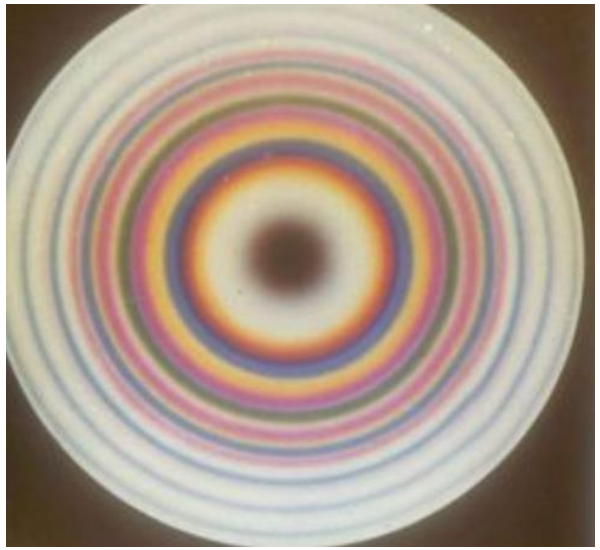




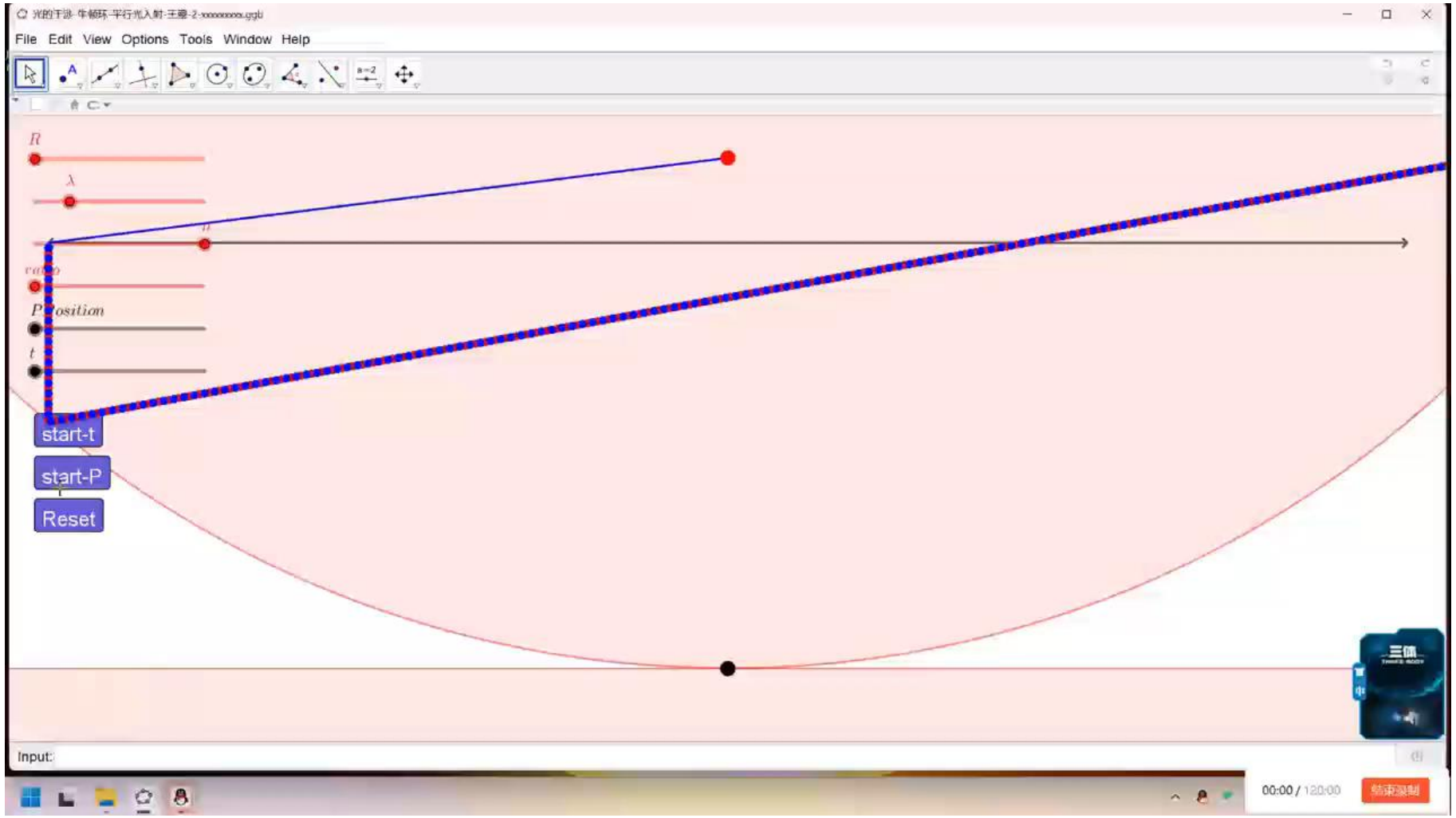


● 等厚干涉的教学动画 （河南科技大学王翠老师制作于2024.10.12~10.13）

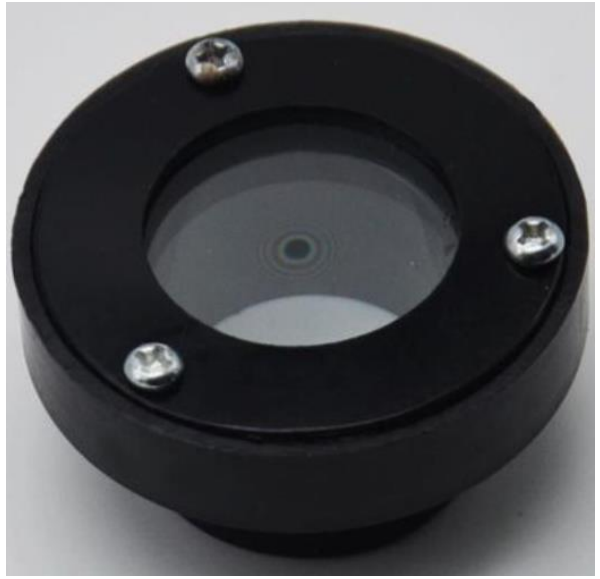


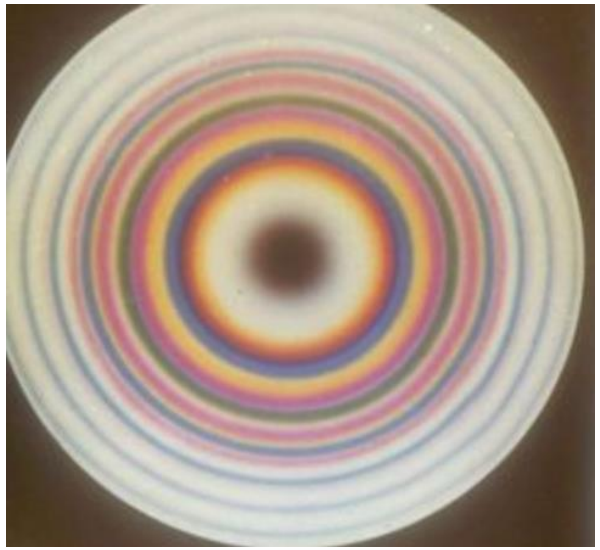


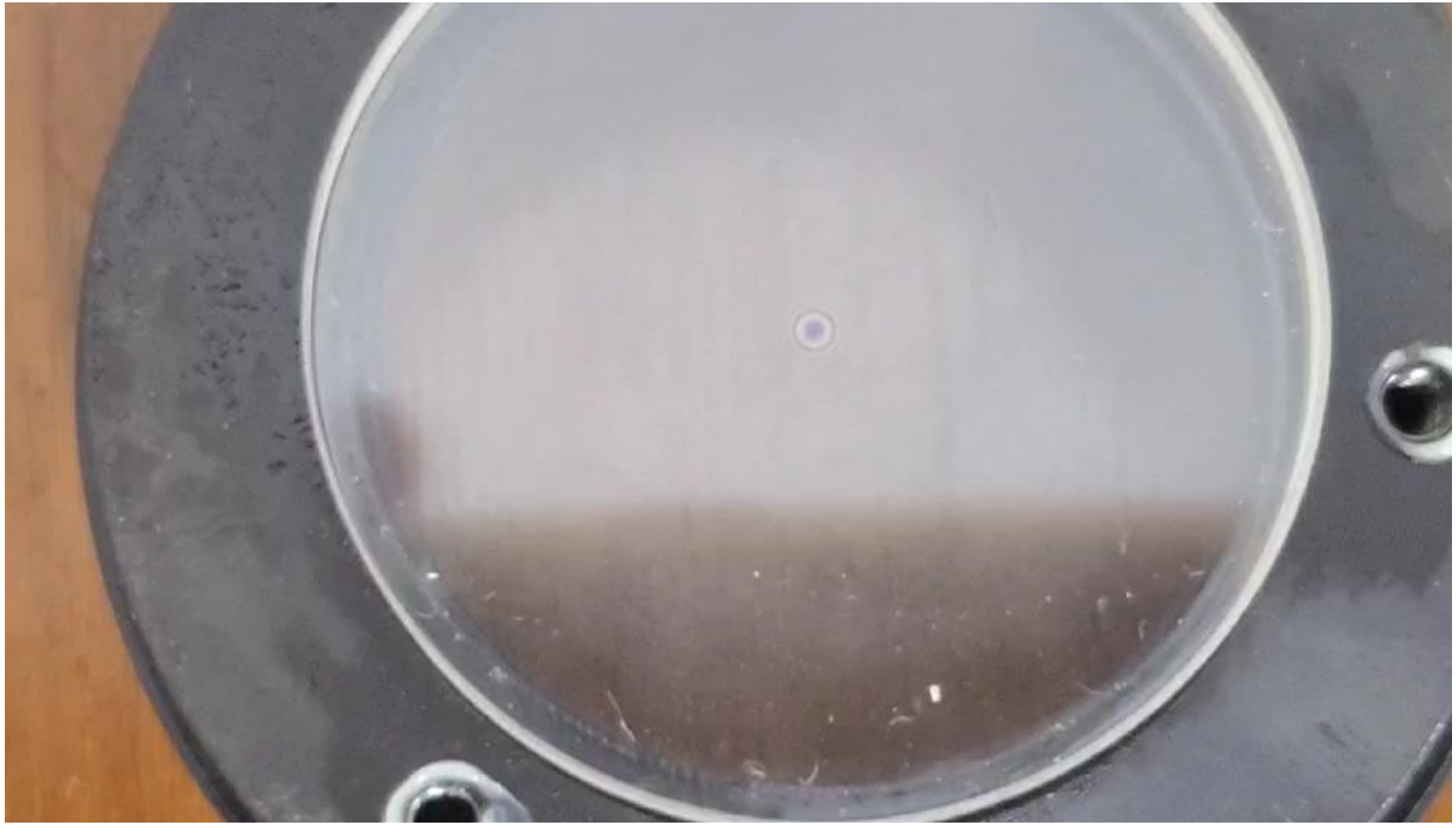
1.2 牛顿环上方的平凸透镜对应的球面半径较小时… …



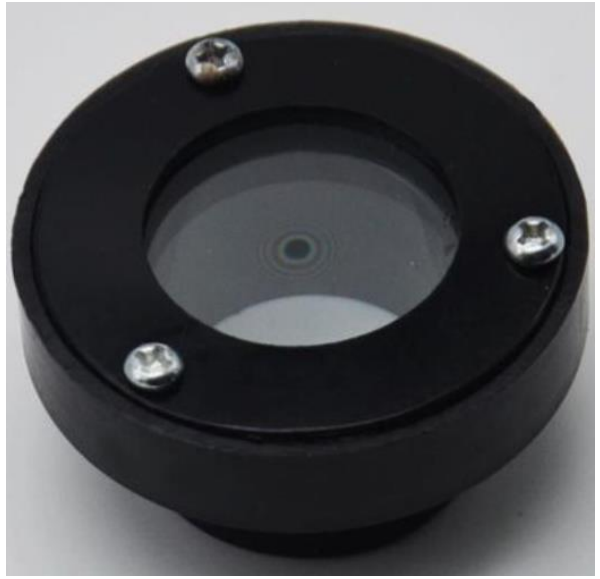
1.3 牛顿环实物录像

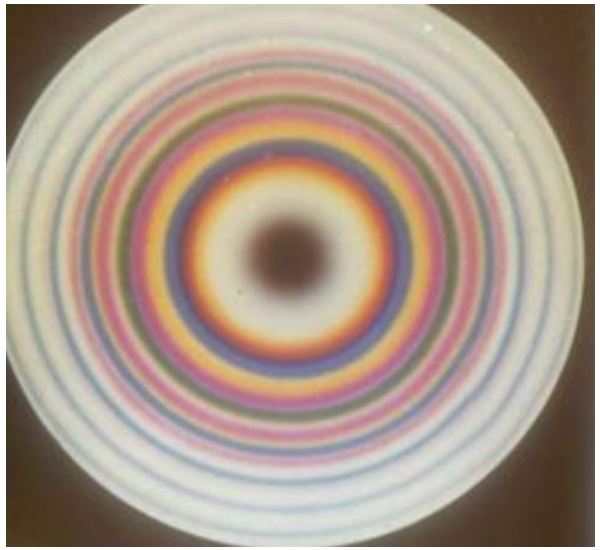






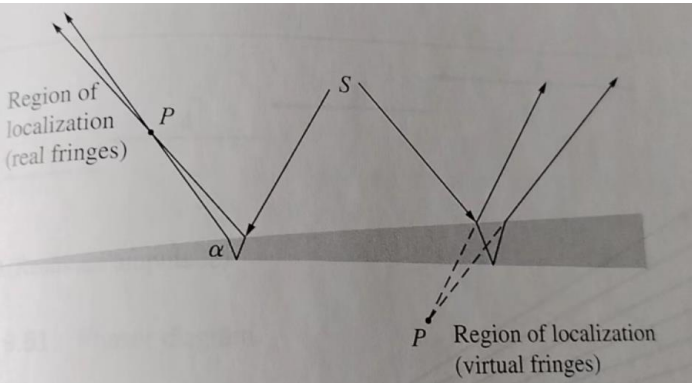
1.4 等倾干涉实物录像对比

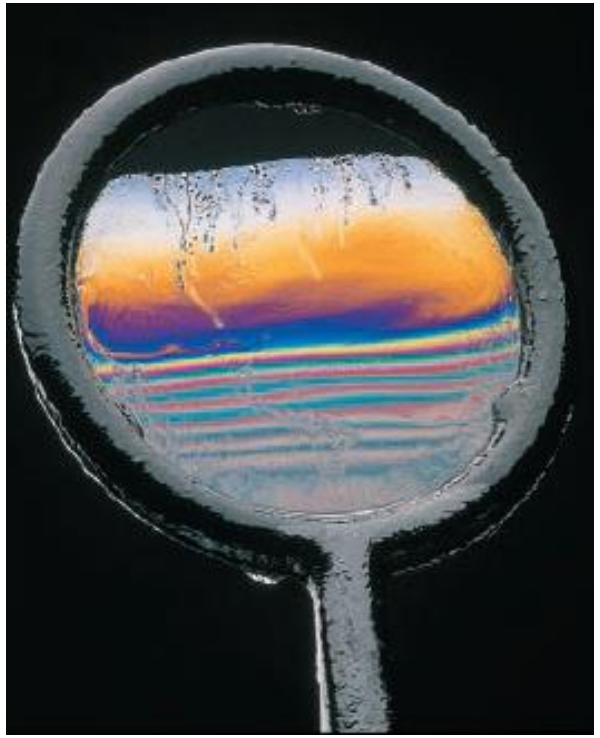


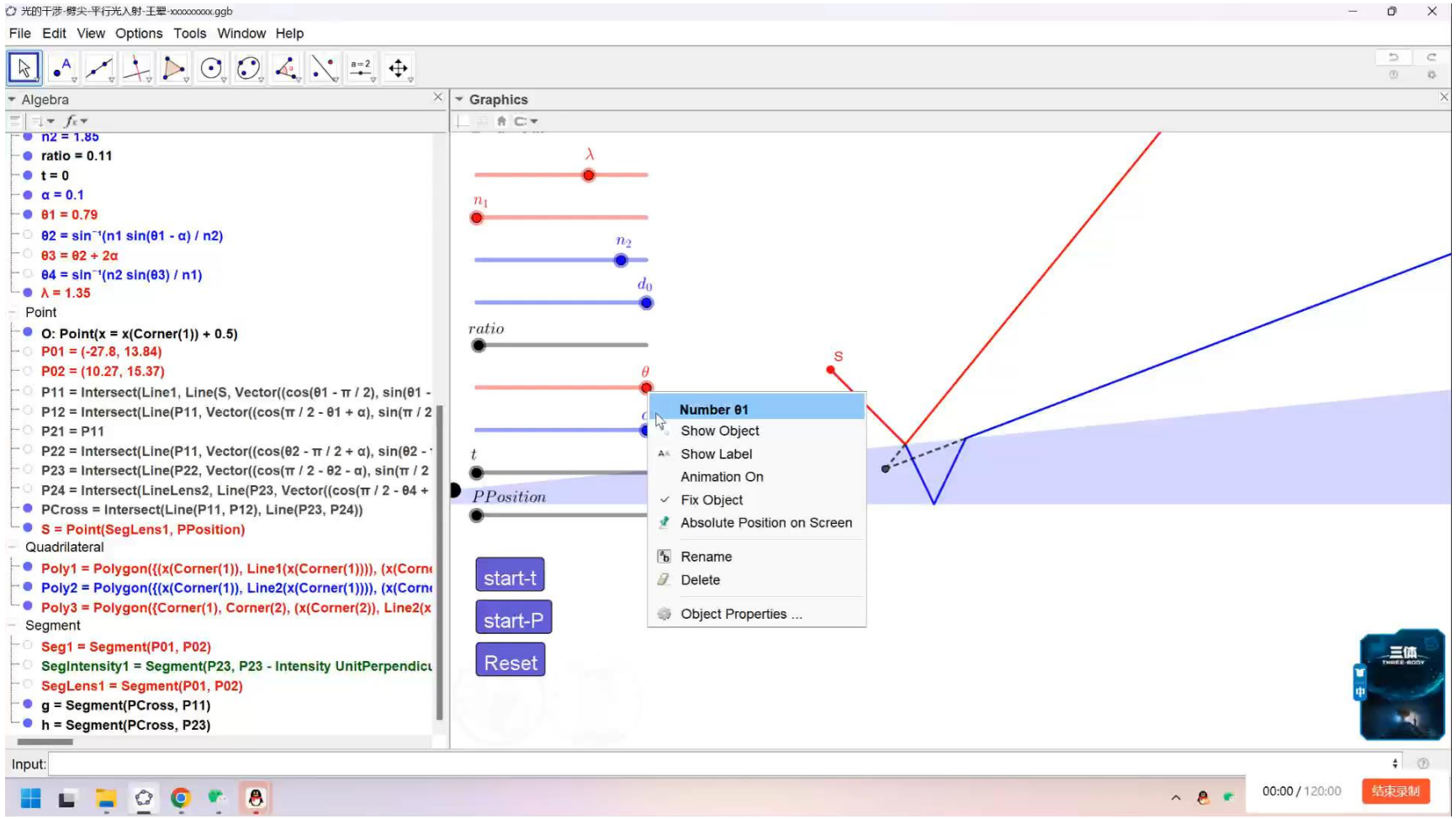


● 等厚干涉的教学动画 （河南科技大学王翟老师制作于2024.10.12~10.13）

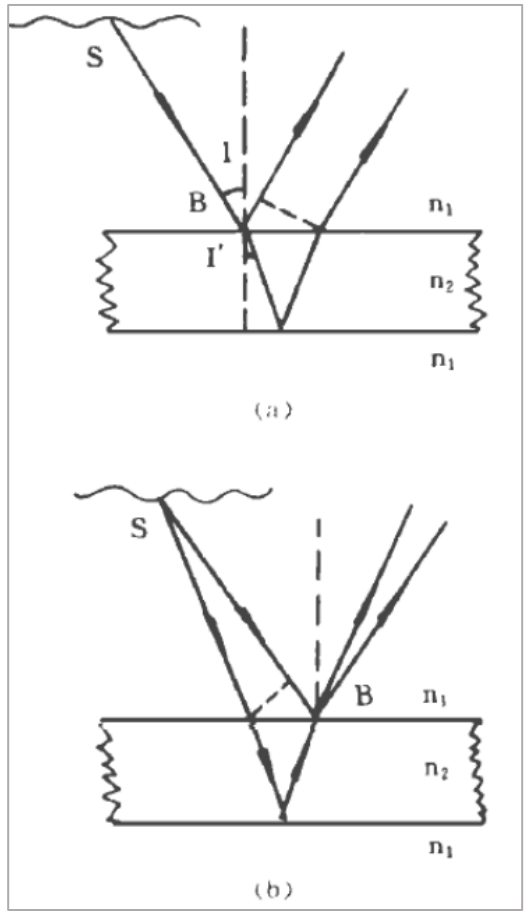
2. 玻璃劈尖 ——干涉条纹的“定域”问题、“虚实”问题



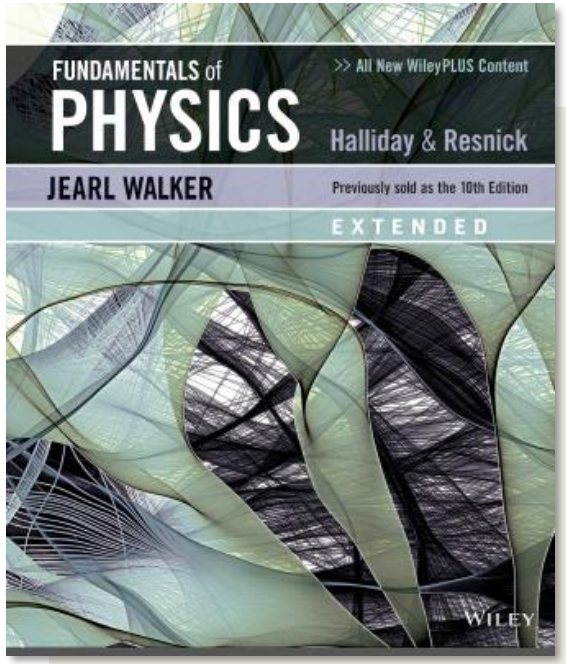




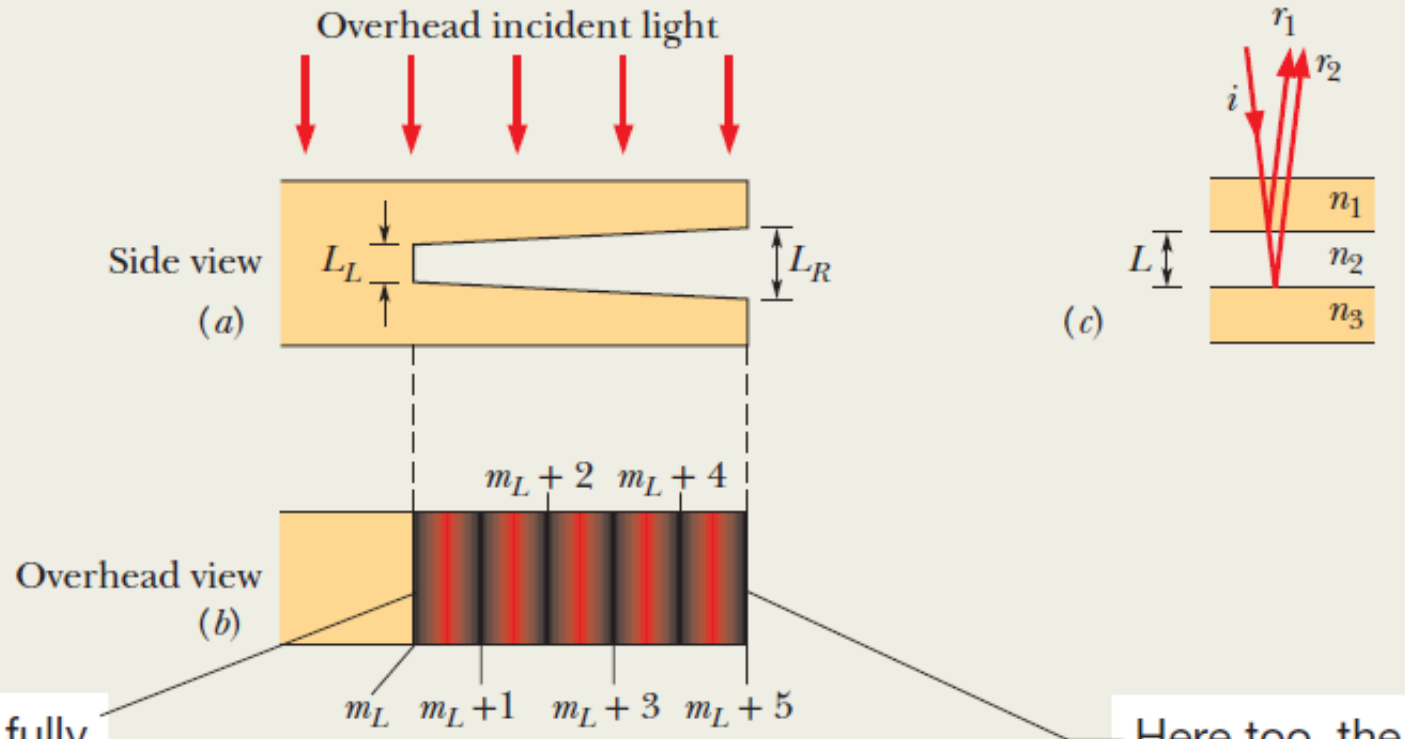
3. 空气劈尖 —— 干涉条纹 “定域”在哪里 .. .. ..



张天波,马建国.薄膜干涉条纹定域和形状问题的讨论[J].安徽机电学院学报(自然科学版),1997,(02):26-29.



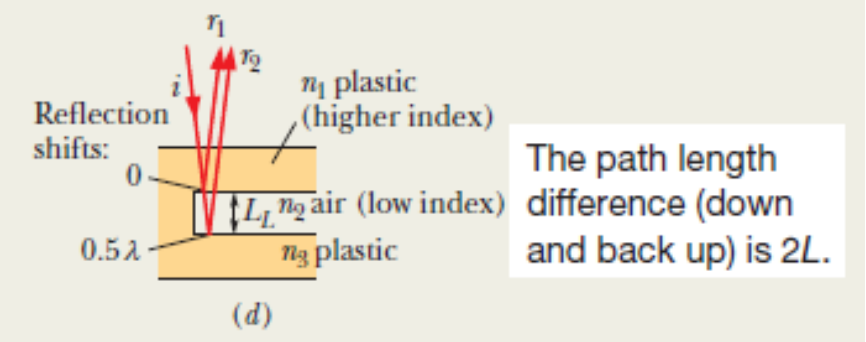
3. 空气劈尖 —— 干涉条纹 “定域”在哪里 .. .. ..

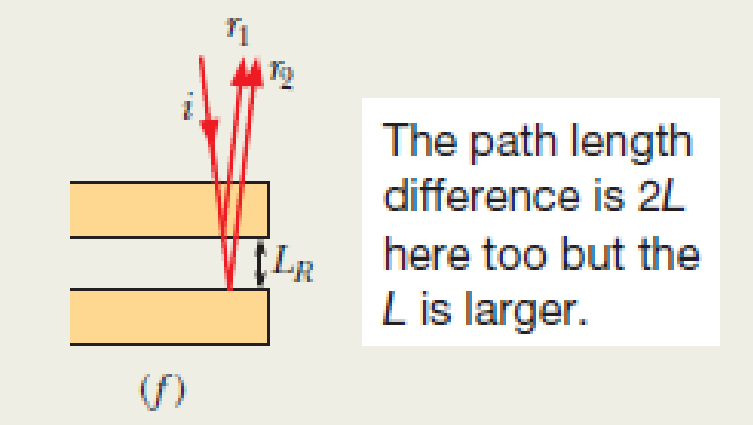


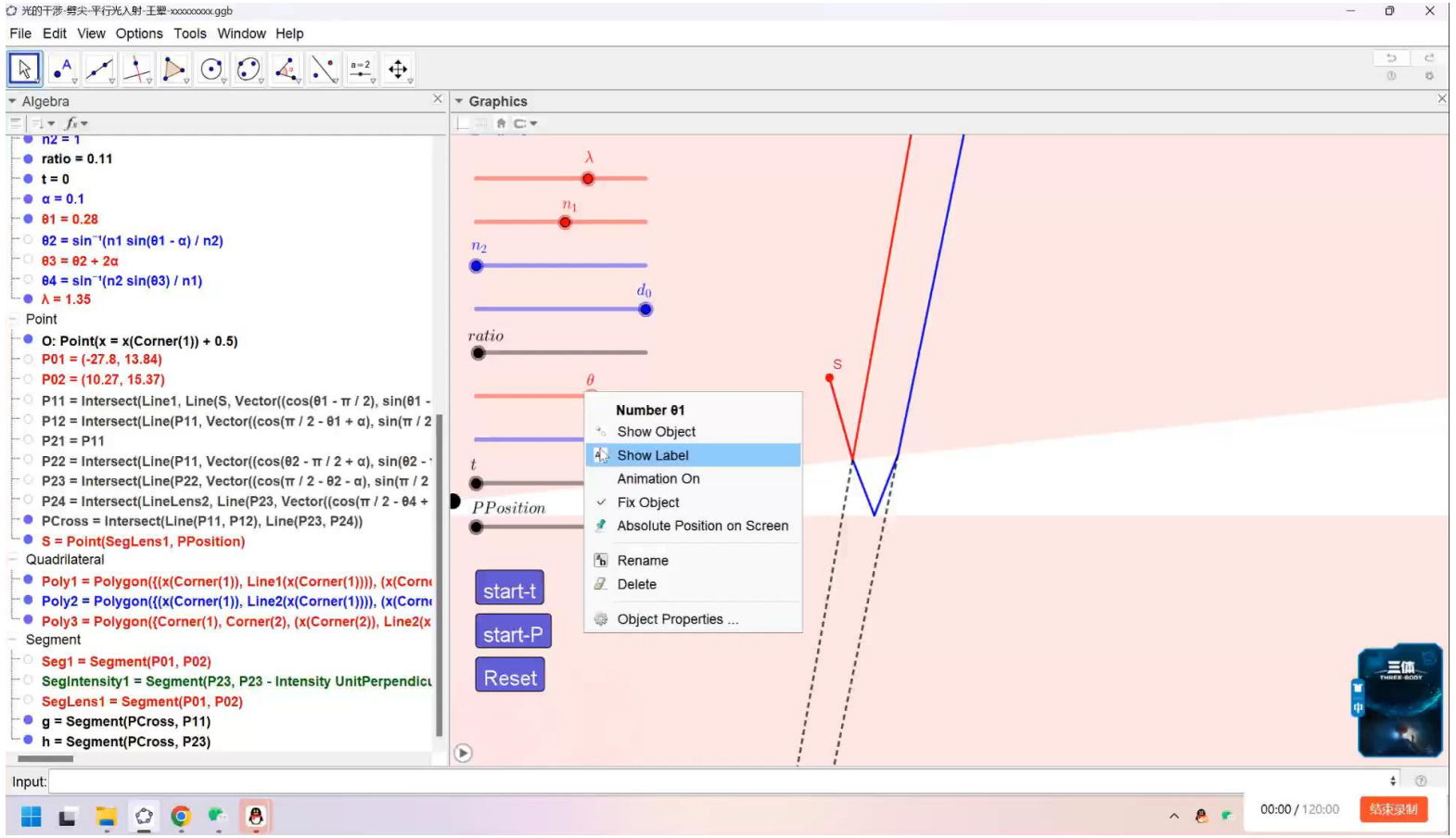
This dark fringe is due to fully destructive interference. So, the reflected rays must be *out* of phase.

Here too, the dark fringe means that the reflected waves are *out* of phase.

3. 空气劈尖 —— 干涉条纹 “定域”在哪里 .. .. ..







谢　谢